

IP Addressing (IPv4)

What is an IP Address?

Definition

An **IP (Internet Protocol) Address** is a **unique logical address** assigned to every device connected to a network. It allows devices to identify and communicate with each other over a network or the Internet.

Think of an IP address as the **home address** of a device. Just as a postal address helps deliver a letter to the correct house, an IP address helps deliver data to the correct device.

Why Do We Need an IP Address?

Without an IP address, devices would not know where to send or receive data.

Benefits

- Uniquely identifies a device on a network
 - Enables communication between devices
 - Helps routers deliver data to the correct destination
 - Makes Internet communication possible
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Real-Life Example

Imagine you want to send a courier to your friend.

The courier company needs your friend's **home address** to deliver the package.

Similarly, when your laptop sends data to a website, it needs the **IP address** of the destination server.

Home Address
↓
Courier Delivery

IP Address
↓
Data Delivery

Types of IP Addresses

There are two versions of IP addresses:

Version	Description
IPv4	32-bit address (Most commonly used)
IPv6	128-bit address (Designed to replace IPv4)

In this chapter, we will focus on **IPv4**.

What is IPv4?

IPv4 (Internet Protocol Version 4) is the most widely used version of the Internet Protocol.

It is a **32-bit logical address** divided into **four parts (octets)**.

Example:

192.168.1.10

Each part is called an **Octet**.

192	168	1	10
↑	↑	↑	↑
Octet1	Octet2	Octet3	Octet4

Range of Each Octet

Each octet contains **8 bits**.

The valid range of each octet is:

0 - 255

Examples of valid IP addresses:

- 192.168.1.10
- 10.0.0.5
- 172.16.100.20

IPv4 Structure

An IPv4 address consists of:

- 32 Bits
- 4 Octets
- Each Octet = 8 Bits

8 Bits 8 Bits 8 Bits 8 Bits

192 . 168 . 1 . 10

Total:

$8 + 8 + 8 + 8 = 32 \text{ Bits}$

Binary Representation

Computers understand only binary (0s and 1s).

Example:

Decimal

192.168.1.10

↓

Binary

11000000.10101000.00000001.00001010

Although humans use decimal numbers, computers process IP addresses in binary.

Network ID and Host ID

An IPv4 address has two parts:

Network ID

Identifies the network.

Host ID

Identifies the device within that network.

Example:

192.168.1.10

A subnet mask determines which portion is the **Network ID** and which is the **Host ID**.

Note: We will study subnet masks in detail in the CIDR and Subnetting chapter.

Rules for a Valid IPv4 Address

A valid IPv4 address must satisfy the following conditions:

- It contains **4 octets**.
 - Each octet ranges from **0 to 255**.
 - Octets are separated by dots (.).
 - No alphabetic characters are allowed.
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Examples

Valid IP Addresses

192.168.1.1
10.0.0.10
172.16.5.100
8.8.8.8

Invalid IP Addresses

300.10.1.5 ❌ (300 is greater than 255)

192.168.1 ❌ (Only three octets)

10.10.abc.5 ❌ (Contains letters)

192.168.1.256 ❌ (256 is greater than 255)

Real-World Example

Suppose your laptop is connected to Wi-Fi.

Laptop
IP Address:
192.168.1.20



Wi-Fi Router
192.168.1.1



Internet

The router uses IP addresses to send and receive data between your laptop and the Internet.

Quick Revision

Feature	Description
Full Form	Internet Protocol Address
Type	Logical Address
IPv4 Size	32 Bits
Number of Octets	4
Range of Each Octet	0 – 255
Format	X.X.X.X

Interview Corner

Q1. What is an IP Address?

Answer:

An IP Address is a unique logical address assigned to a device that enables communication over a network.

Q2. What is the full form of IP?

Answer:

Internet Protocol.

Q3. How many bits are there in an IPv4 address?

Answer:

32 Bits.

Q4. How many octets are there in an IPv4 address?

Answer:

4 Octets.

Q5. What is the valid range of each octet?

Answer:

0 to 255.

Q6. Is an IP Address a physical address or a logical address?

Answer:

It is a **Logical Address**.

Q7. Give an example of a valid IPv4 address.

Answer:

192.168.1.10

Key Takeaways

- An IP Address uniquely identifies a device on a network.
- IPv4 is a **32-bit logical address**.
- It consists of **4 octets**, each containing **8 bits**.
- The valid range of each octet is **0–255**.
- Routers use IP addresses to deliver data to the correct destination.